

Object Oriented Programming

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Last Week

- Un-named temporary return
- Operator overloading
 - Prefix vs postfix
 - Arithmetic assignment
 - Subscript operator
- Friend functions

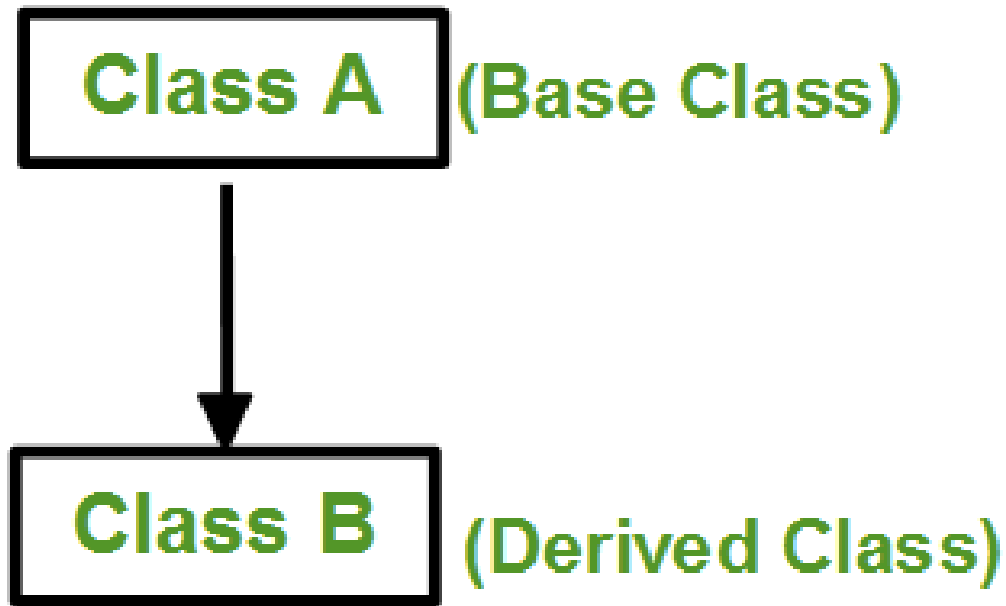
This Week (and Next weeks)

- Friend function in class diagram???
- Inheritance
 - Motivation
 - Class diagram
 - Example

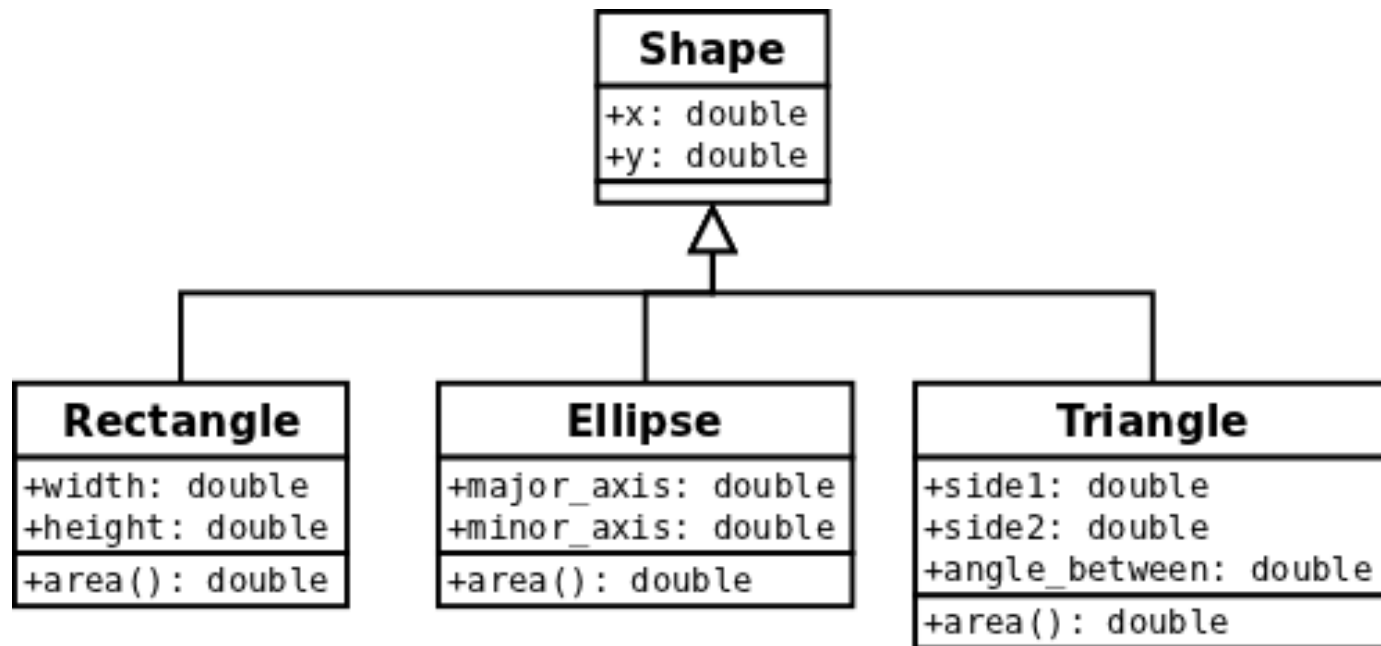
Motivation

- Real world examples
 - Shape (square, circle, triangle ...)
 - Person (Dean, HoD, Lecturer, Student)
 - Vehicle (Bus, Truck, Car ...)

Class Diagram



Shape Class Diagram



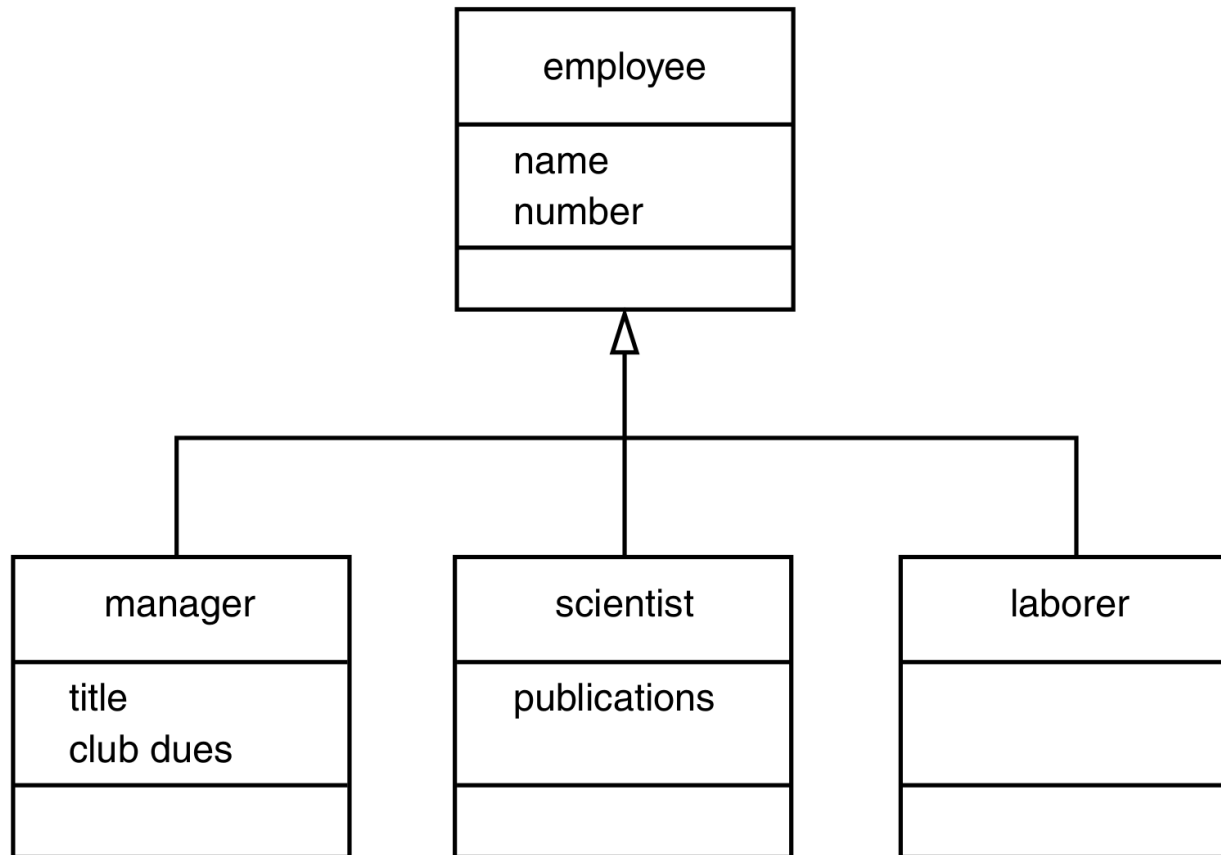
Example

```
class A {  
    int data;  
public:  
    void f(int arg) { data = arg; }  
    int g() { return data; }  
};  
  
class B : public A { };  
  
int main() {  
    B obj;  
    obj.f(20);  
    cout << obj.g() << endl;  
}
```

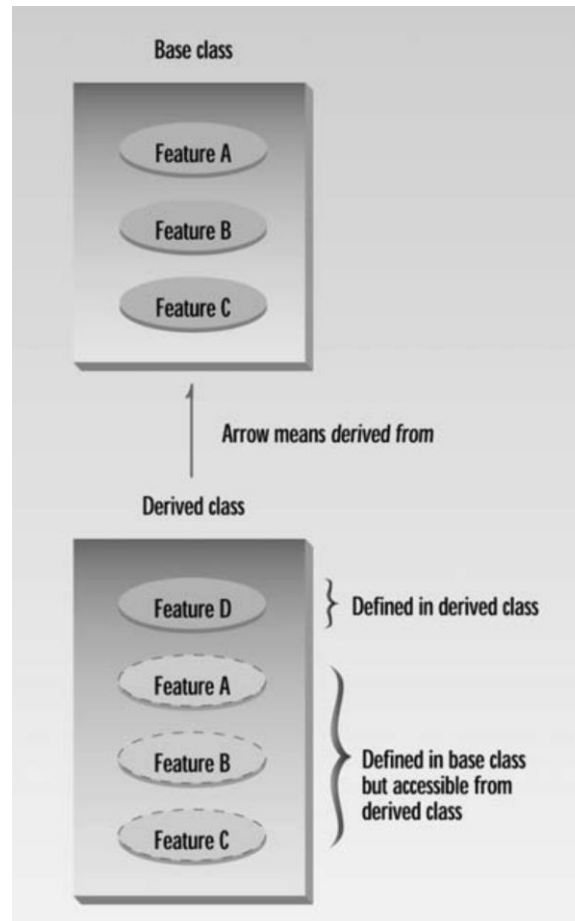
Quiz: What will be the output?

```
5 class Mother {
6     public:
7         Mother ()
8             { cout << "Mother: no parameters\n"; }
9         Mother (int a)
10            { cout << "Mother: int parameter\n"; }
11 };
12
13 class Daughter : public Mother {
14     public:
15         Daughter (int a)
16             { cout << "Daughter: int parameter\n\n"; }
17 };
18
19 class Son : public Mother {
20     public:
21         Son (int a) : Mother (a)
22             { cout << "Son: int parameter\n\n"; }
23 };
24
25 int main () {
26     Daughter kelly(0);
27     Son bud(0);
28
29     return 0;
30 }
```

Quiz: Model the following inheritance in C++



Illustration



Counter example

- From Counter to CountdownCounter
- How should we deal with constructors?

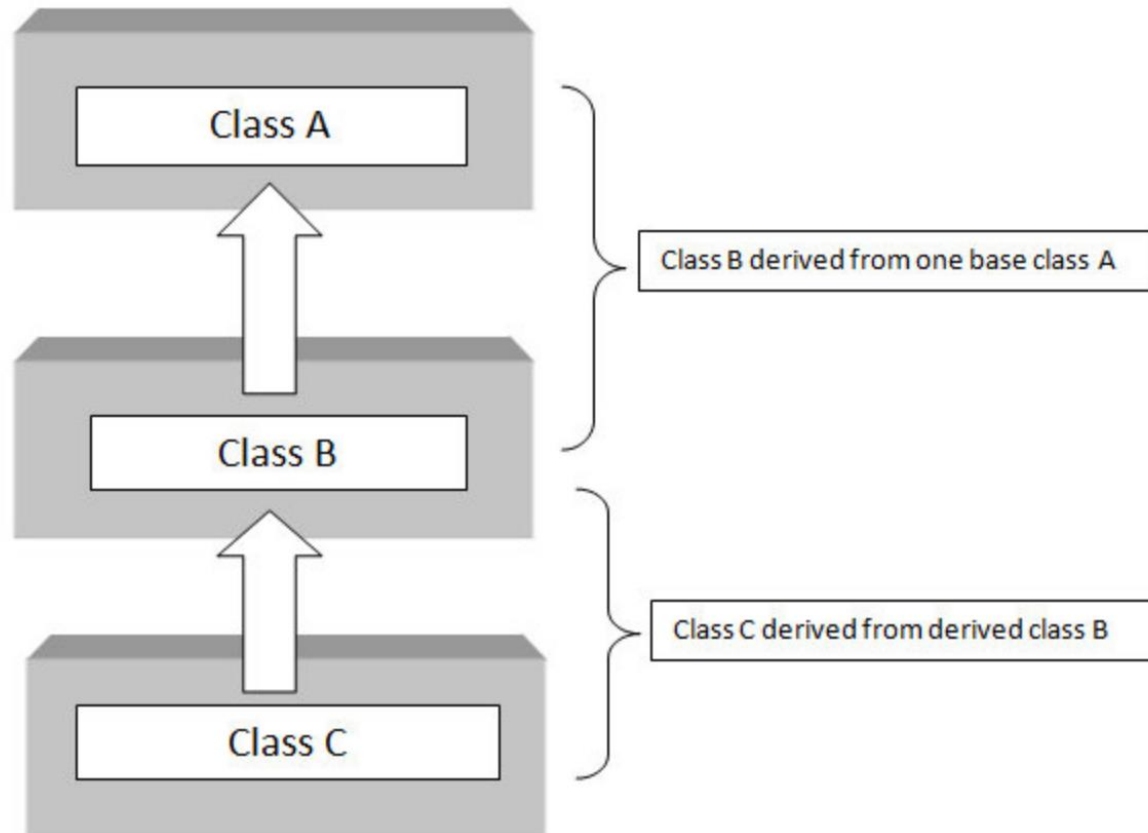
Derived Class Constructors

- `CountDn() : Counter() { }`
- `CountDn(int c) : Counter(c) { }`

Stack to Safe Stack example

- Safe stack should be derived from original stack
- How do push() and pop() in Stack2 access push() and pop() in Stack?

Multiple Inheritance



Summary

- Inheritance is an important C++ concept